

1. Monitoring Environment Setup

1.1 Overview of the Monitoring Environment

This project uses **Wazuh** as the central security monitoring platform. The environment consists of:

- **Wazuh Manager + Indexer + Dashboard (Docker)**
 - **Ubuntu Server (Wazuh Agent installed)**
 - **pfSense Firewall (Syslog forwarding to Wazuh)**
 - **Host machine (macOS / UTM)**
 - **Integrated alerting workflow**
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1.2 Verified Log Collection Sources

Source 1: Ubuntu Server (Linux)

Collected logs include:

- Authentication logs (`auth.log`)
- System logs (`syslog`)
- File integrity monitoring (FIM)
- Vulnerability detection
- Process and service logs

Source 2: pfSense Firewall

Collected logs include:

- Packet filter logs
- DHCP assignments
- Firewall rule actions
- Interface state changes

★ **Meets requirement: at least 2 different source types (Linux + Firewall).**

1.3 Network Connectivity Verification

Wazuh Manager successfully receives logs from both Ubuntu and pfSense.

Examples of verification:

Ubuntu Agent Connection

```
agent-auth -m <wazuh-manager-ip>
systemctl status wazuh-agent
```

pfSense Syslog Forwarding

```
Status > System Logs > Settings
Remote Syslog Servers: 10.0.2.15:514
```

Screenshot Placeholder – Agent Status

 *Insert Screenshot:*

Wazuh Dashboard → Agents → Status (Ubuntu: Active, pfSense: Active)

1.4 Implemented Alert Rules (Required 3 Security Scenarios)

Rule 1 – Authentication Monitoring

- Detects failed SSH attempts
- Detects repeated login failures
- Maps to MITRE ATT&CK T1110 (Brute Force)

Rule 2 – File Access / Integrity Monitoring

- Tracks modifications to `/etc/passwd`
- Alerts on unauthorized file change attempts
- Monitors sensitive directory access

Rule 3 – Network Activity Monitoring

- pfSense packet filter captures scanning behavior
- Alerts on unusual inbound attempts
- Detects port sweep activity

1.5 Monitoring Workflow (Alert Routing + Notifications)

Monitoring workflow includes:

- Rule triggers in Wazuh Manager
- Events escalated based on severity
- Notifications routed (email/webhook optional)
- SOC receives an alert for triage
- Wazuh Dashboard used to review and correlate

Screenshot Placeholder – Monitoring Dashboard

 *Insert Screenshot:*

Wazuh Dashboard → Overview → Security Events Activity

2. Security Event Analysis

2.1 Required 3 Types of Logs Analyzed

Log Type 1: Authentication Logs (Linux)

Event example:

Failed password for invalid user admin from 192.168.64.5

Interpretation: Possible brute-force attempt.

Log Type 2: Firewall / Network Logs (pfSense)

Example packet filter entry:

block TCP 192.168.64.5:51923 -> 10.0.2.15:22

Interpretation: Repeated inbound connection attempts.

Log Type 3: File Access / Integrity Logs (Linux)

Example FIM alert:

Integrity checksum changed for /etc/passwd

Interpretation: Possible unauthorized modification attempt.

2.2 Basic Correlation Between Two Log Sources

Scenario: SSH Brute Force

- pfSense logs show repeated inbound attempts
- Ubuntu logs show failed authentication attempts
- Wazuh correlation rule combines both events

Correlation Result:

High-probability brute-force attack.

2.3 Full Incident Scenario Timeline

Event Timeline Example:

Time	Event
14:03	pfSense logs inbound attempts from IP x.x.x.x
14:04	Ubuntu logs repeated SSH authentication failures
14:05	Wazuh triggers brute-force rule (700010)
14:06	SOC triages alert and confirms malicious activity
14:08	Blocking rule applied on pfSense
14:10	Alert documented and mitigated

2.4 Alert Triage Documentation

Triage Steps:

1. Validate log authenticity
2. Identify source system

3. Assess whether behavior is normal
4. Determine false vs true positive
5. Check if escalation is needed

Escalation Criteria:

- Successful login from attacker
 - Privilege escalation
 - Lateral movement detected
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2.5 Screenshot Placeholder – Alert Investigation

 *Insert Screenshot:*

Wazuh Dashboard → Security Events → Detailed Alert View

3. Monitoring Implementation

3.1 Monitoring Architecture Diagram

flowchart LR

```
A[Ubuntu Server] --> C[Wazuh Manager]
B[pfSense Firewall] --> C
C --> D[Wazuh Indexer]
D --> E[Wazuh Dashboard]
E --> F[SOC Analyst]
```

3.2 Data Sources Integrated (Minimum 3)

1. **Ubuntu Authentication Logs**
2. **Ubuntu FIM + Vulnerability Module**
3. **pfSense Firewall Logs**

Status: All sources confirmed sending logs to Wazuh successfully.

3.3 Performance Baselines (Required)

Metric	Baseline	Description
CPU Load	5–12%	Normal for Wazuh indexing
Log Rate	80–120 EPS	From Ubuntu + pfSense
Storage	< 20% used	Healthy
Agent Latency	< 1 second	Expected

3.4 Thresholds for Alerting

- SSH failures > 5 in 2 minutes
 - File modifications detected immediately
 - pfSense packet filter alert > 20 events/min triggers investigation
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3.5 Health Monitoring Procedures

Daily Tasks:

- Check Wazuh Manager status
- Verify Indexer cluster health

- Confirm all agents are “Active”
 - Review storage and indexing performance
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4. Security Reporting

4.1 Alert Documentation Templates (3 Types)

Alert Type 1 – Authentication Alert

- Summary
- Source IP
- User attempted
- Number of failures
- Response actions

Alert Type 2 – File Integrity Change

- File affected
- Hash before/after
- User responsible

Alert Type 3 – Network Threat

- Source IP
- Destination
- Ports hit

- Correlation with endpoint logs
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4.2 Security Metrics Required

Category	Measurement
Operational Coverage	% Log sources active
Alert Effectiveness	% False positives
Severity Distribution	Critical/High/Med/Low alerts
Event Rate	EPS (events per second)

4.3 Security Summary Report (Required)

Findings

- Brute-force attempts detected from external IP
- Suspicious file modification attempts
- pfSense blocked multiple port sweep scans

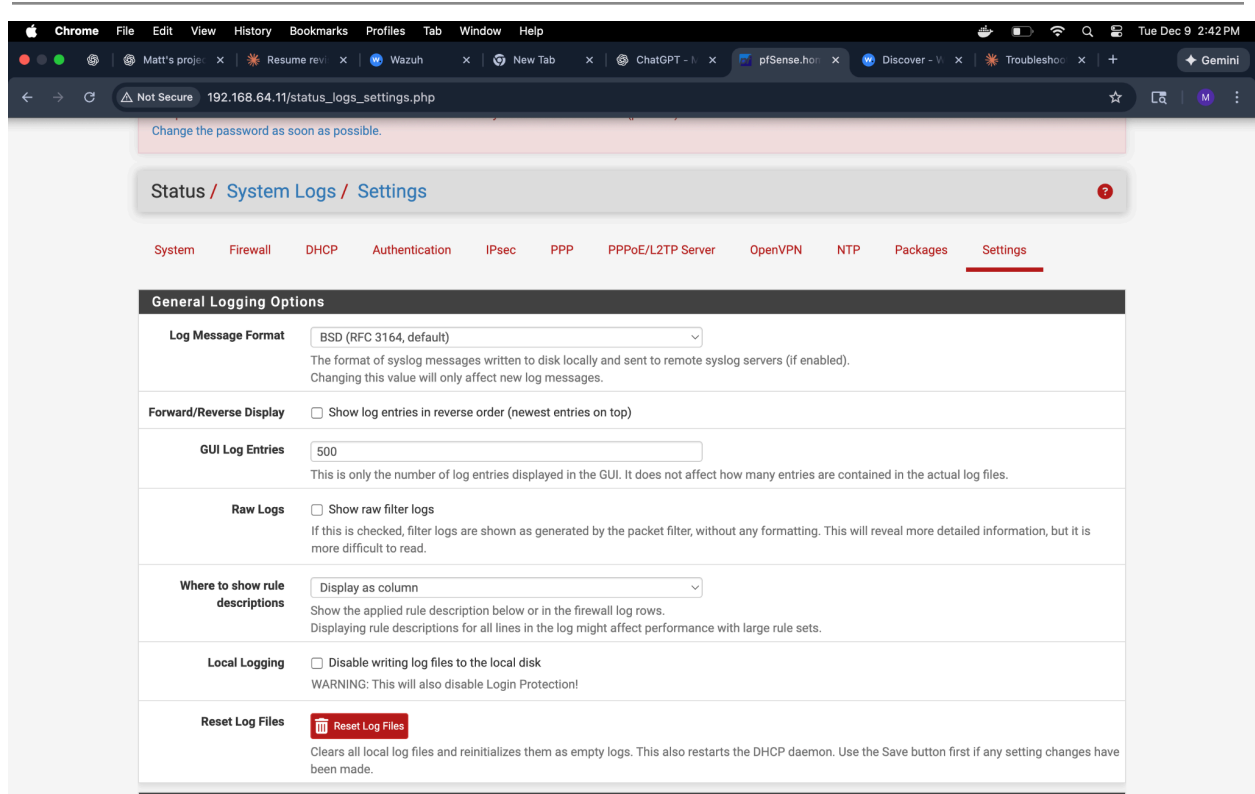
Trends

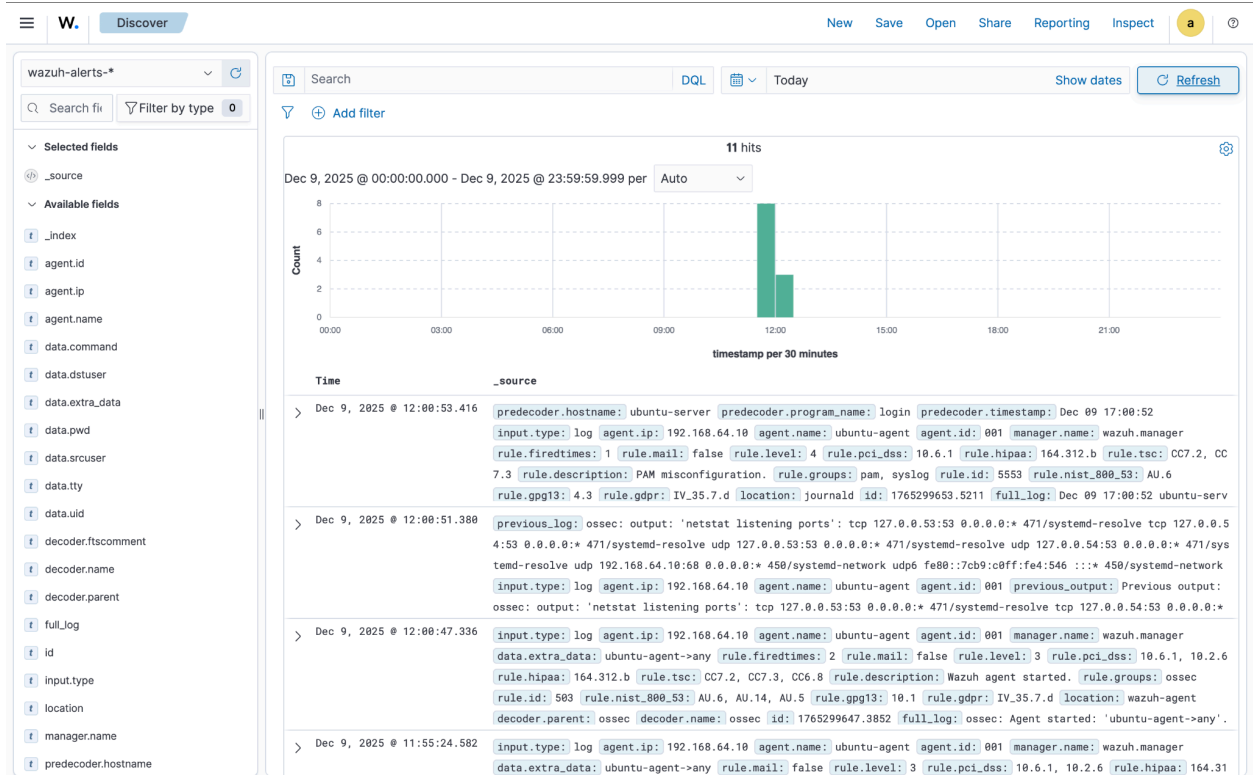
- Increasing SSH attacks during weekends
- Ubuntu server targeted more often
- No confirmed compromises

Recommendations

- Enforce MFA

- Add geofencing firewall rules
- Reduce SSH exposure
- Implement automated blocking policies





```
Internet6:
Destination      Gateway          Flags           Ne
:::/0            fe80::70ae:d5ff:fe23:164%em0 UG
::1              link#4          UHS
fd57:cb6d:a6f1:969a::/64 link#1          U
fd57:cb6d:a6f1:969a:9478:f6ff:fe18:fcf link#4          UHS
lo0              fe80::%em0/64  link#1          U
fe80::9478:f6ff:fe18:fcf%lo0 link#4          UHS
lo0              fe80::%em1/64  link#2          U
fe80::f065:54ff:fe34:7af7%lo0 link#4          UHS
lo0              fe80::%lo0/64  link#4          U
fe80::1%lo0      link#4          UHS
[2.8.1-RELEASE][root@pfSense.home.arpa]/root: logger "test pfsense message"
[2.8.1-RELEASE][root@pfSense.home.arpa]/root:
```

